

CAK2HAB3 - Dasar Kecerdasan Artifisial

Reasoning: Propositional Logic

Lecturer Team





Five types of Logic

Type of logic	Reference to real world	Type of knowledge known by Agent
Propositional Logic	Facts	True / False / Unknown
First-order Logic	Facts, Object, Relation	True / False / Unknown
Temporal Logic	Facts, Object, Relation, Time	True / False / Unknown
Probability Theory	Facts	Degree of belief [0, 1] (certainty factor)
Fuzzy Logic	Truth values (degree of truth)	Confidence level [0, 1] (certainty factor)



Propositional Logic



Propositional Logic

- ▶ The simplest logic
- ▶ Very easy to understand
- ▶ Makes it easier to distinguish reasoning techniques by searching techniques.
- ▶ "Propositional logic is not the study of truth, but of the relationship between the truth of one statement and that of another" (Hedman 2004).



Backus-Naur Form (BNF)

Sentence $\rightarrow$ *AtomicSentence* | *ComplexSentence*

AtomicSentence $\rightarrow$ *True* | *False*
| *P* | *Q* | *R* | ...

ComplexSentence $\rightarrow$ (*Sentence*)
| *Sentence* *Connective* *Sentence*
| $\neg$ *Sentence*

Connective $\rightarrow$ $\wedge$ | $\vee$ | $\Leftrightarrow$ | $\Rightarrow$



Truth Tables for 5-Connectives

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
		negation	conjunction	disjunction	implication	Biconditional
F	F	T	F	F	T	T
F	T	T	F	T	T	F
T	F	F	F	T	F	F
T	T	F	T	T	T	T



Inference Rule

1. **Modus Ponens** or **Implication-Elimination**:

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$

2. **And-Elimination**:

$$\frac{\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n}{\alpha_i}$$

3. **And-Introduction**:

$$\frac{\alpha_1, \alpha_2, \dots, \alpha_n}{\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n}$$

4. **Or-Introduction**:

$$\frac{\alpha_i}{\alpha_1 \vee \alpha_2 \vee \dots \vee \alpha_n}$$

5. **Double-Negation-Elimination**:

$$\frac{\neg \neg \alpha}{\alpha}$$

6. **Unit Resolution**:

$$\frac{\alpha \vee \beta, \neg \beta}{\alpha}$$

7. **Resolution**:

$$\frac{\alpha \vee \beta, \neg \beta \vee \gamma}{\alpha \vee \gamma}$$

Equivalent with

$$\frac{\neg \alpha \Rightarrow \beta, \beta \Rightarrow \gamma}{\neg \alpha \Rightarrow \gamma}$$

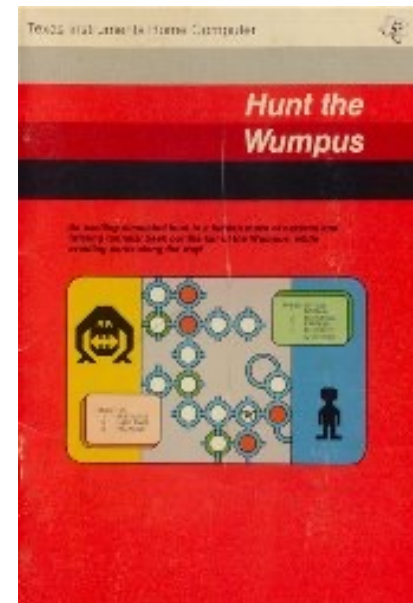


The Wumpus World



Hunt the Wumpus

- ▶ an early computer game,
- ▶ based on a simple hide and seek format featuring a mysterious monster (the Wumpus) that lurks deep inside a network of rooms.
- ▶ originally developed as a text-based game written in BASIC.



The Wumpus World

- A simple world (as is the Block World) for which to represent knowledge and to reason
- An earlier simulation environment in which knowledge-based agents can show their worth
- Introduced by Genesereth, and is discussed in Russell-Norvig Artificial intelligence book,
- Inspired by 1972 video game Hunt the Wumpus



The Wumpus World

4				
3		  		
2				
1				
	1	2	3	4



The Wumpus World

- The **agent** explores a cave consisting of rooms connected by passageways. (4x4 grid room)
- Lurking somewhere in the cave is the **Wumpus**, a beast that eats any agent that enters its room.
- Some rooms contain bottomless **pits** that trap any agent that wanders into the room.
- Occasionally, there is a heap of **gold** in a room.
- The goal is to collect the gold and exit the world without being eaten

Wumpus World: Agent / Actuators

- Agent Movements (Actuators)
 - **Turn right, Turn Left,**
 - Move **Forward** at current facing direction
 - Perceive Bump if it hits a wall
 - **Grab** gold, **Release** (drop) Gold
 - **Shoot** the only arrow at current facing direction
 - The arrow will fly through the rooms until either it hits the Wumpus or hits a wall
 - **Climb** out the cave (only at square[1,1])



Wumpus World: Environment

WUMPUS

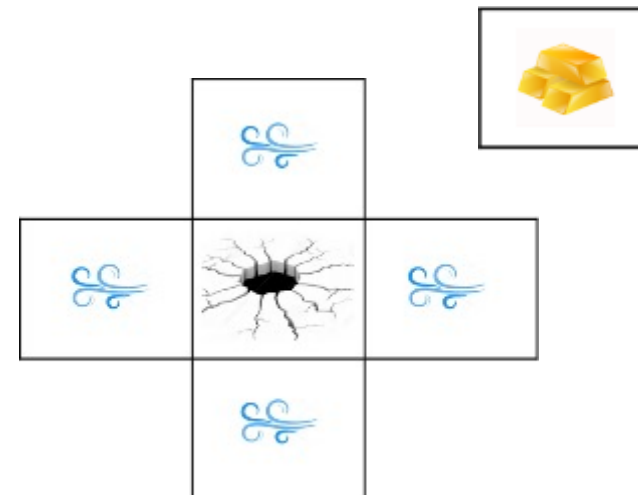
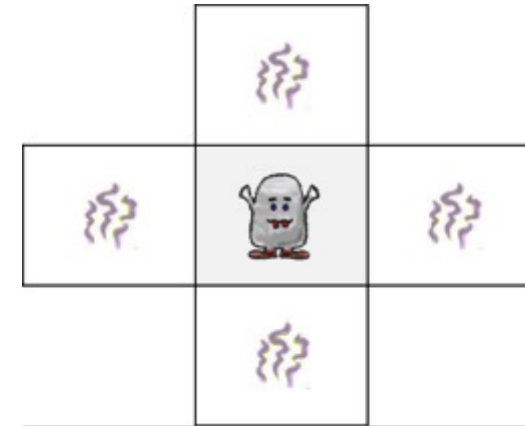
- Agent dies if it enters a square with a live Wumpus in it
- Squares adjacent to Wumpus are smelly (**STENCH**)
- Wumpus will **SCREAM** If it is killed by an arrow

3 PITs

- Agent dies if it enters a Pit
- Squares adjacent to Pit are **BREEZY**

GOLD

- **GLITTER** iff Gold is in the same square



Wumpus World: Sensors

- Five item ordered vector
 - Perceive the status of current square [**STENCH**, **BREEZE**, **GLITTER**, **BUMP**, **SCREAM**]
- The agent perceives
 - a **STENCH** in the square containing the Wumpus and in the adjacent squares (not diagonally)
 - a **BREEZE** in the squares adjacent to a pit
 - a **GLITTER** in the square where the gold is
 - a **BUMP**, if it walks into a wall
 - a woeful **SCREAM** everywhere in the cave, if the Wumpus is killed
- The agent cannot perceive its own location



Wumpus World: Performance Measure

- ▶ Start:
 - Start at square $[1,1]$, 1 Arrow
- ▶ Goal:
 - find the gold, return it to the start square, leave the cave
- ▶ Points
 - +1000 points reward for climbing out w/ the gold
 - -1 point deducted for every action taken
 - -10000 points penalty for getting killed
 - -10 for using up the arrow





Wumpus World Characterization

- ▶ Fully Observable? **No**
 - only local perception
- ▶ Deterministic? **Yes**
 - outcomes exactly specified
- ▶ Episodic? **No**
 - sequential at the level of actions
- ▶ Static? **Yes**
 - Wumpus and Pits do not move
- ▶ Discrete? **Yes**
- ▶ Single-agent? **Yes**
 - Wumpus is essentially a natural feature



Wumpus World using Propositional Logic



Knowledge-based System

- Agent performs an action based on the perceptual reasoning of his Knowledge Based.
- At the beginning of the game, there is no fact at all in KB because Agent has not received any percept.
- KB only contains some rules (rule) which is knowledge of the environment
- To perform the proper action, agents must be equipped with rules to translate knowledge into action.

Knowledge-based System

```
function PL-WUMPUS-AGENT(percept) returns an action
  inputs: percept, a list, [stench, breeze, glitter]
  static: KB, a knowledge base, initially containing the "physics" of the wumpus world
         x, y, orientation, the agent's position (initially [1,1]) and orientation (initially right)
         visited, an array indicating which squares have been visited, initially false
         action, the agent's most recent action, initially null
         plan, an action sequence, initially empty

  update x, y, orientation, visited based on action
  if stench then TELL(KB,  $S_{x,y}$ ) else TELL(KB,  $\neg S_{x,y}$ )
  if breeze then TELL(KB,  $B_{x,y}$ ) else TELL(KB,  $\neg B_{x,y}$ )
  if glitter then action  $\leftarrow$  grab
  else if plan is nonempty then action  $\leftarrow$  POP(plan)
  else if for some fringe square [i,j], ASK(KB, ( $\neg P_{i,j} \wedge \neg W_{i,j}$ )) is true or
         for some fringe square [i,j], ASK(KB, ( $P_{i,j} \vee W_{i,j}$ )) is false then do
    plan  $\leftarrow$  A*-GRAPH-SEARCH(ROUTE-PROBLEM([x,y], orientation, [i,j], visited))
    action  $\leftarrow$  POP(plan)
  else action  $\leftarrow$  a randomly chosen move
  return action
```



S	Stench		P	Pit		A	Agent		R	Right
B	Breeze		G	Gold		F	Forward		C	Climb
W	Wumpus		V	Visited		L	Left		G	Grab Gold

General Knowledge

► Known facts:

$$\begin{aligned} W_{x,y} &\Rightarrow S_{x,y} \wedge S_{x-1,y} \wedge S_{x,y-1} \wedge S_{x+1,y} \wedge S_{x,y+1}, & x, y = 1, \dots, 4 \\ P_{x,y} &\Rightarrow B_{x-1,y} \wedge B_{x,y-1} \wedge B_{x+1,y} \wedge B_{x,y+1}, & x, y = 1, \dots, 4 \\ \neg W_{x,y} \wedge \neg P_{x,y} &\Rightarrow \text{Save}_{x,y}, & x, y = 1, \dots, 4 \end{aligned}$$

► Therefore

$$\begin{aligned} B_{x,y} &\Leftrightarrow P_{x-1,y} \vee P_{x,y-1} \vee P_{x+1,y} \vee P_{x,y+1}, & x, y = 1, \dots, 4 \\ S_{x,y} &\Leftrightarrow W_{x-1,y} \vee W_{x,y-1} \vee W_{x+1,y} \vee W_{x,y+1} & x, y = 1, \dots, 4 \end{aligned}$$



S	Stench		P	Pit		A	Agent		R	Right
B	Breeze		G	Gold		F	Forward		C	Climb
W	Wumpus		V	Visited		L	Left		G	Grab Gold

General Knowledge

Wumpus location

$$W_{1,1} \vee W_{1,2} \vee \dots \vee W_{4,4}$$

$$\neg W_{x,y} \vee \neg W_{x',y'}, \quad x, y, x', y' = 1, \dots, 4, x \neq x' \text{ or } y \neq y'$$

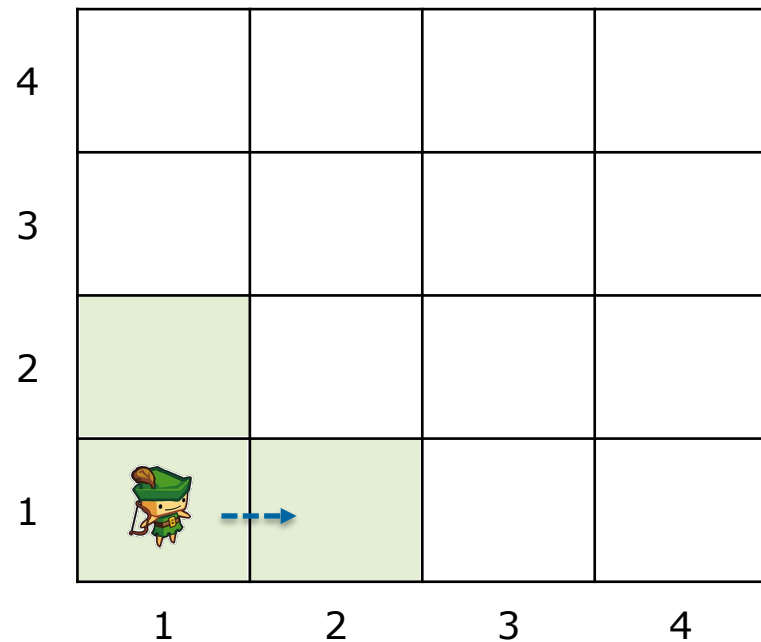
Movement Translation

$$T_1: A_{x,y} \wedge FacingR_A \wedge P_{x+1,y} \Rightarrow \neg F$$

$$T_2: A_{x,y} \wedge FacingU_A \wedge W_{x,y+1} \Rightarrow \neg F$$

...

The Wumpus World



[None, None, None, None, None]

Action:

- R, F

Knowledge:

- $\neg S_{1,1} \wedge \neg B_{1,1} \wedge \neg G_{1,1}$
- $\neg W_{1,1} \wedge \neg P_{1,1}$
- $Save_{1,1}$
- $\neg W_{1,2} \wedge \neg W_{2,1}$
- $\neg P_{1,2} \wedge \neg P_{2,1}$
- $Save_{1,2} \wedge Save_{2,1}$

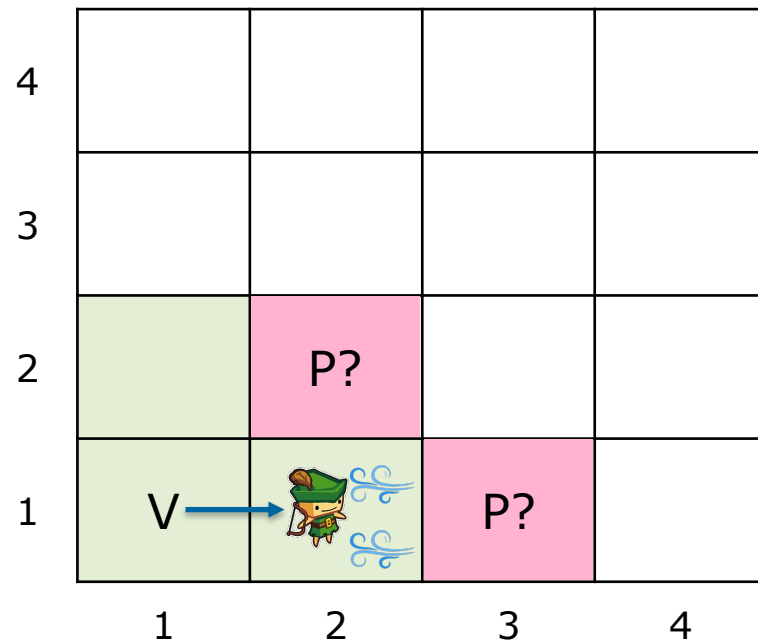
Inference:

- $\neg W_{1,1} \wedge \neg P_{1,1} \Rightarrow Save_{1,1}$
- $\neg S_{1,1} \Rightarrow \neg W_{1,2} \wedge \neg W_{2,1}$
- $\neg B_{1,1} \Rightarrow \neg P_{1,2} \wedge \neg P_{2,1}$
- $\neg W_{1,2} \wedge \neg P_{1,2} \Rightarrow Save_{1,2}$
- $\neg W_{2,1} \wedge \neg P_{2,1} \Rightarrow Save_{2,1}$

and-elimination,
unit resolution



The Wumpus World



[None, BREEZE, None, None, None]

Action:

- R, R, F, R, F

Knowledge:

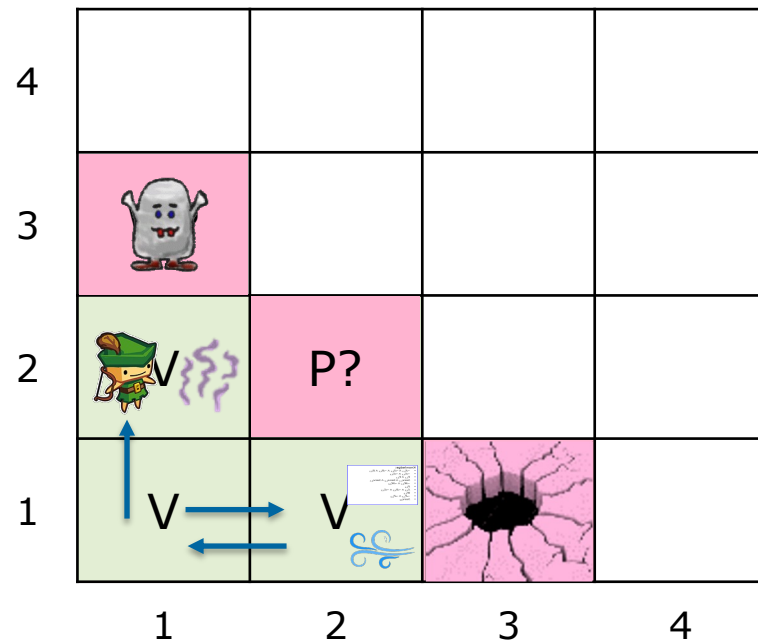
- $\neg S_{1,1} \wedge \neg B_{1,1} \wedge \neg G_{1,1} \wedge V_{1,1}$
- $Save_{1,1} \wedge Save_{1,2} \wedge Save_{2,1}$
- $\neg S_{2,1} \wedge B_{2,1} \wedge \neg G_{2,1}$
- $\neg W_{3,1} \wedge \neg W_{2,2}$
- $P_{3,1} \vee P_{2,2}$

Inference:

- $\neg S_{2,1} \Rightarrow \neg W_{3,1} \wedge \neg W_{2,2}$
- $B_{2,1} \Leftrightarrow P_{3,1} \vee P_{2,2}$

and-elimination,
unit resolution,

The Wumpus World



[STENCH, None, None, None, None]

Action:

- R, F

Knowledge:

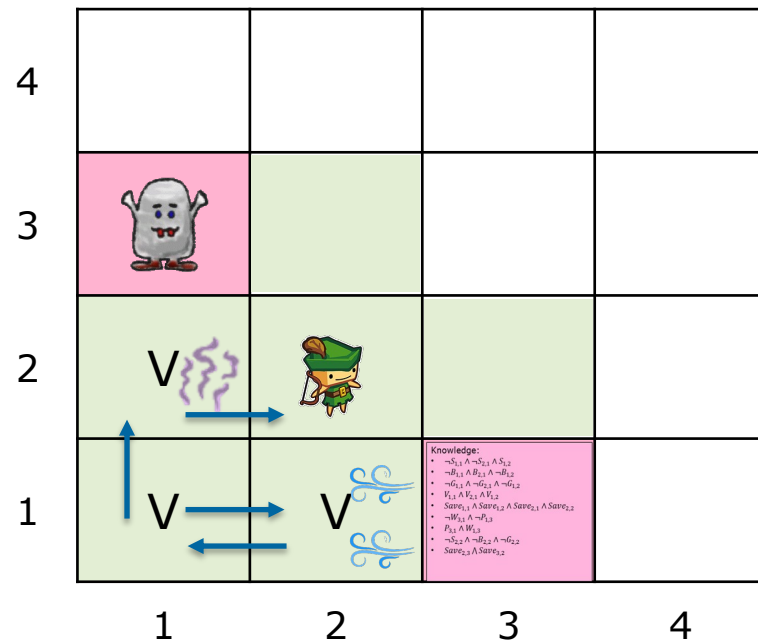
- $\neg S_{1,1} \wedge \neg S_{2,1} \wedge \neg B_{1,1} \wedge B_{2,1}$
- $\neg G_{1,1} \wedge \neg G_{2,1}$
- $V_{1,1} \wedge V_{2,1}$
- $Save_{1,1} \wedge Save_{1,2} \wedge Save_{2,1}$
- $\neg W_{3,1} \wedge \neg W_{2,2}$
- $P_{3,1} \vee P_{2,2}$
- $S_{1,2} \wedge \neg B_{1,2} \wedge \neg G_{1,2}$
- $W_{1,3}$
- $\neg P_{1,3} \wedge \neg P_{2,2}$
- $Save_{2,2}$

Inference:

- $S_{1,2} \Leftrightarrow W_{1,3} \vee W_{2,2}$
- $(W_{1,3} \vee W_{2,2}) \wedge \neg W_{2,2} \vdash W_{1,3}$
- $\neg B_{1,2} \Rightarrow \neg P_{1,3} \wedge \neg P_{2,2}$
- $(P_{3,1} \vee P_{2,2}) \wedge \neg P_{2,2} \vdash P_{3,1}$
- $\neg W_{2,2} \wedge \neg P_{2,2} \Rightarrow Save_{2,2}$

disjunctive syllogism

The Wumpus World



[None, None, None, None, None]

Action:

- F

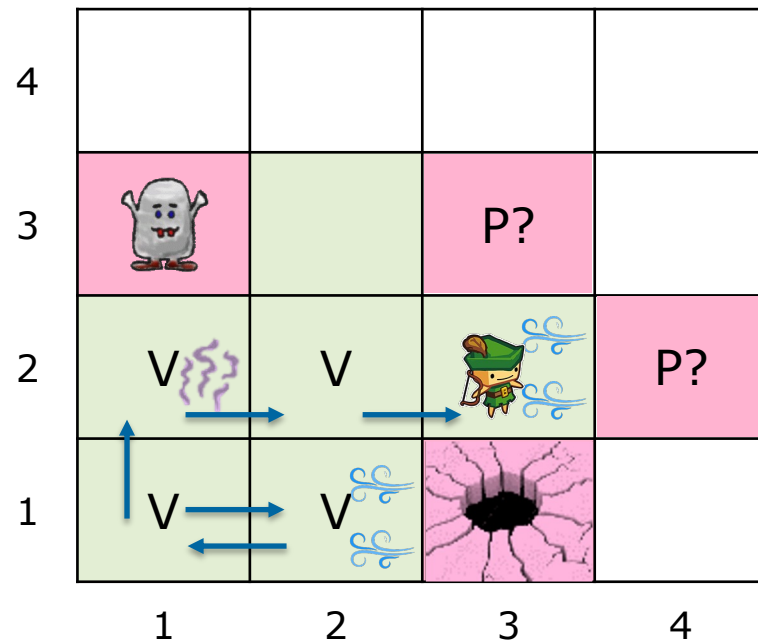
Knowledge:

- $\neg S_{1,1} \wedge \neg S_{2,1} \wedge S_{1,2}$
- $\neg B_{1,1} \wedge B_{2,1} \wedge \neg B_{1,2}$
- $\neg G_{1,1} \wedge \neg G_{2,1} \wedge \neg G_{1,2}$
- $V_{1,1} \wedge V_{2,1} \wedge V_{1,2}$
- $Save_{1,1} \wedge Save_{1,2} \wedge Save_{2,1} \wedge Save_{2,2}$
- $\neg W_{3,1} \wedge \neg P_{1,3}$
- $P_{3,1} \wedge W_{1,3}$
- $\neg S_{2,2} \wedge \neg B_{2,2} \wedge \neg G_{2,2}$
- $Save_{2,3} \wedge Save_{3,2}$

Inference:

- $\neg S_{2,2} \Rightarrow \neg W_{2,3} \wedge \neg W_{3,2}$
- $\neg B_{2,2} \Rightarrow \neg P_{2,3} \wedge \neg P_{3,2}$
- $\neg W_{2,3} \wedge \neg P_{2,3} \Rightarrow Save_{2,3}$
- $\neg W_{3,2} \wedge \neg P_{3,2} \Rightarrow Save_{3,2}$

The Wumpus World



[None, BREEZE, None, None, None]

Action:

- R, R, F, R, F

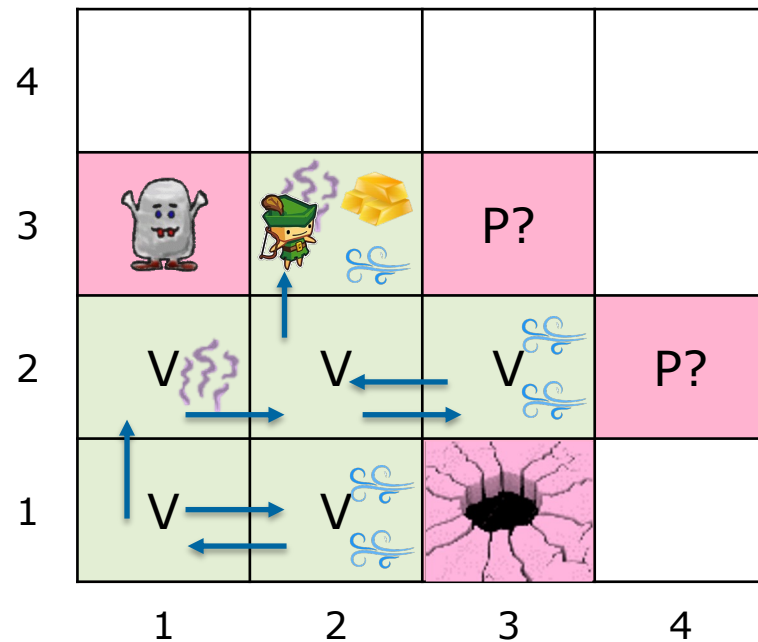
Knowledge:

- $\neg S_{1,1} \wedge \neg S_{2,1} \wedge S_{1,2} \wedge \neg S_{2,2}$
- $\neg B_{1,1} \wedge B_{2,1} \wedge \neg B_{1,2} \wedge \neg B_{2,2}$
- $\neg G_{1,1} \wedge \neg G_{2,1} \wedge \neg G_{1,2} \wedge \neg G_{2,2}$
- $V_{1,1} \wedge V_{2,1} \wedge V_{1,2} \wedge V_{2,2}$
- $Save_{1,1} \wedge Save_{1,2} \wedge Save_{2,1} \wedge Save_{2,2}$
- $Save_{2,3} \wedge Save_{3,2}$
- $\neg W_{3,1} \wedge \neg P_{1,3}$
- $P_{3,1} \wedge W_{1,3}$
- $\neg S_{3,2} \wedge B_{3,2} \wedge \neg G_{3,2}$
- $\neg W_{3,3} \wedge \neg W_{4,2}$
- $P_{3,3} \vee P_{4,2}$

Inference:

- $\neg S_{3,2} \Rightarrow \neg W_{3,3} \wedge \neg W_{4,2}$
- $B_{3,2} \Rightarrow P_{3,3} \vee P_{4,2}$

The Wumpus World



[STENCH, BREEZE, GLITTER, None, None]

Action:

- G, L, L, F, R, F, L, F, C

Knowledge:

- $\neg S_{1,1} \wedge \neg S_{2,1} \wedge S_{1,2} \wedge \neg S_{2,2} \wedge \neg S_{3,2}$
- $\neg B_{1,1} \wedge B_{2,1} \wedge \neg B_{1,2} \wedge \neg B_{2,2} \wedge B_{3,2}$
- $\neg G_{1,1} \wedge \neg G_{2,1} \wedge \neg G_{1,2} \wedge \neg G_{2,2} \wedge \neg G_{3,2}$
- $V_{1,1} \wedge V_{2,1} \wedge V_{1,2} \wedge V_{2,2} \wedge V_{3,2}$
- $Save_{1,1} \wedge Save_{1,2} \wedge Save_{2,1} \wedge Save_{2,2}$
- $Save_{2,3} \wedge Save_{3,2}$
- $\neg W_{3,1} \wedge \neg P_{1,3} \wedge \neg W_{3,3} \wedge \neg W_{4,2}$
- $P_{3,1} \wedge W_{1,3}$
- $P_{3,3} \vee P_{4,2}$
- $S_{2,3} \wedge B_{2,3} \wedge G_{2,3}$

Inference:

- $G_{2,3} \Rightarrow Grab(G), Out$



Pros and cons of propositional logic

- (+) Propositional logic is declarative
- (+) Propositional logic allows partial/disjunctive/negated information
 - (unlike most data structures and databases)
- (+) Propositional logic is compositional:
 - meaning of $P \wedge Q$ is derived from meaning of P and of Q
- (+) Meaning in propositional logic is context-independent
 - (unlike natural language, where meaning depends on context)



Pros and cons of propositional logic

- (-) Propositional logic has very limited expressive power
 - (unlike natural language)
 - E.g., cannot say "pits cause breezes in adjacent squares"
 - except by writing one sentence for each square
- Propositional logic only deals with "facts", statements that may or may not be true of the world, e.g. "It is raining".



The natural language words may have slightly different meanings

- ▶ Example:
 - $A \wedge B$ and $B \wedge A$ should always have the same meaning.

- ▶ But the sentences
 - “She became sick and she went to the doctor.”
 - and
 - “She went to the doctor and she became sick.”might have different meanings.



Question?





THANK YOU